

Web Technologies Lab – 04

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Aim:

To develop a React-based Online Shopping Product Catalog that demonstrates dynamic searching, filtering, and conditional rendering using React state management. Features Search products by name. Filter by category (Electronics, Clothing, Grocery). Filter by price range. Show "Out of Stock" if quantity is 0. Display product count. Concepts useState Multiple filters filter() map() Conditional rendering

Source Code

File 1: index.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Product Catalog</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/main.jsx"></script>
  </body>
</html>
```

File 2: main.jsx

```
import { StrictMode } from "react";
```

```
import { createRoot } from "react-dom/client";
import ProductCatalog from "../product_catalog.jsx";

createRoot(document.getElementById("root")).render(
  <StrictMode>
    <ProductCatalog />
  </StrictMode>
);
```

File 3: product_catalog.jsx

```
import { useState } from "react";

const PRODUCTS = [
  { id: 1, name: "Wireless Headphones", category: "Electronics", price: 2499, quantity: 12 },
  { id: 2, name: "Bluetooth Speaker", category: "Electronics", price: 1799, quantity: 0 },
  { id: 3, name: "Smart Watch", category: "Electronics", price: 3999, quantity: 5 },
  { id: 4, name: "Cotton T-Shirt", category: "Clothing", price: 599, quantity: 30 },
  { id: 5, name: "Denim Jacket", category: "Clothing", price: 1999, quantity: 0 },
  { id: 6, name: "Running Shoes", category: "Clothing", price: 2799, quantity: 8 },
  { id: 7, name: "Basmati Rice 5kg", category: "Grocery", price: 549, quantity: 40 },
  { id: 8, name: "Olive Oil 1L", category: "Grocery", price: 899, quantity: 15 },
  { id: 9, name: "Almonds 500g", category: "Grocery", price: 649, quantity: 0 },
  { id: 10, name: "Laptop Stand", category: "Electronics", price: 1299, quantity: 20 },
  { id: 11, name: "Mechanical Keyboard", category: "Electronics", price: 3499, quantity: 7 },
  { id: 12, name: "Wool Sweater", category: "Clothing", price: 2199, quantity: 14 },
  { id: 13, name: "Leather Belt", category: "Clothing", price: 899, quantity: 0 },
  { id: 14, name: "Canvas Sneakers", category: "Clothing", price: 1699, quantity: 22 },
  { id: 15, name: "Green Tea 250g", category: "Grocery", price: 349, quantity: 50 },
  { id: 16, name: "Honey 500g", category: "Grocery", price: 429, quantity: 0 },
];

const CATEGORIES = ["All", "Electronics", "Clothing", "Grocery"];
const PRICE_RANGES = [
  { label: "Any price", min: 0, max: Infinity },
  { label: "Under ₹1000", min: 0, max: 999 },
  { label: "₹1000 – ₹2500", min: 1000, max: 2500 },
  { label: "Above ₹2500", min: 2501, max: Infinity },
];
```

```
];
```

```
// Palette: Cream #E5DECA, Ocean Blue #7E9DA2, Avocado #898433, Olive #45441A, Dark Green #282C15.
```

```
const CATEGORY_THEME = {
  Electronics: { bg: "#dbe6e8", accent: "#282c15", tag: "#b9cdd1" },
  Clothing: { bg: "#e8e4c3", accent: "#45441a", tag: "#d3cd93" },
  Grocery: { bg: "#dedbc4", accent: "#282c15", tag: "#c3c19a" },
};
```

```
export default function ProductCatalog() {
  const [search, setSearch] = useState("");
  const [category, setCategory] = useState("All");
  const [rangeIndex, setRangeIndex] = useState(0);
```

```
  const range = PRICE_RANGES[rangeIndex];
```

```
  const filtered = PRODUCTS.filter((p) => {
    const matchesSearch = p.name.toLowerCase().includes(search.toLowerCase());
    const matchesCategory = category === "All" || p.category === category;
    const matchesPrice = p.price >= range.min && p.price <= range.max;
    return matchesSearch && matchesCategory && matchesPrice;
  });
```

```
  return (
    <div style={styles.page}>
      <div style={styles.wrap}>
        <header style={styles.header}>
          <h1 style={styles.title}>Product Catalog</h1>
          <p style={styles.subtitle}>Search, filter, and browse the store</p>
        </header>

        <div style={styles.controls}>
          <input
            style={styles.input}
            type="text"
            placeholder="Search products by name..."
            value={search}
            onChange={(e) => setSearch(e.target.value)}
          />
        </div>
      </div>
    </div>
  );
}
```

```

/>
<select
  style={styles.select}
  value={category}
  onChange={(e) => setCategory(e.target.value)}
>
  {CATEGORIES.map((c) => (
    <option key={c} value={c}>{c}</option>
  ))}
</select>
<select
  style={styles.select}
  value={rangeIndex}
  onChange={(e) => setRangeIndex(Number(e.target.value))}
>
  {PRICE_RANGES.map((r, i) => (
    <option key={r.label} value={i}>{r.label}</option>
  ))}
</select>
</div>

<p style={styles.count}>
  Showing <strong>{filtered.length}</strong>{" "}
  {filtered.length === 1 ? "product" : "products"}
</p>

{filtered.length === 0 ? (
  <p style={styles.empty}>No products match your filters. Try widening the search.</p>
) : (
  <div style={styles.grid}>
    {filtered.map((p) => {
      const theme = CATEGORY_THEME[p.category];
      return (
        <div key={p.id} style={{ ...styles.card, background: theme.bg }}>
          <div style={styles.cardTop}>
            <span style={{ ...styles.tag, background: theme.tag, color: theme.accent }}>
              {p.category}
            </span>
            {p.quantity === 0 && <span style={styles.outOfStock}>Out of Stock</span>}
          </div>
        </div>
      );
    })}
  </div>

```

```

    </div>
    <h3 style={styles.name}>{p.name}</h3>
    <p style={{ ...styles.price, color: theme.accent }}>
      ₹{p.price.toLocaleString("en-IN")}
    </p>
    <p style={styles.qty}>
      {p.quantity > 0 ? ` ${p.quantity} in stock` : "Currently unavailable"}
    </p>
  </div>
  );
  }}}
</div>
)}
</div>
</div>
);
}

```

```

const styles = {
  page: {
    minHeight: "100vh",
    background: "linear-gradient(180deg, #e5deca 0%, #c7cdc3 50%, #a3b0ab 100%)",
    padding: "40px 48px",
    fontFamily: "'Times New Roman', Times, serif",
  },
  wrap: { maxWidth: "100%", margin: "0 auto" },
  header: { marginBottom: 32 },
  title: { margin: 0, fontSize: 40, fontWeight: 700, letterSpacing: -0.5, color: "#282c15" },
  subtitle: { margin: "8px 0 0", fontSize: 18, color: "#45441a" },
  controls: { display: "flex", flexWrap: "wrap", gap: 14, marginBottom: 20 },
  input: { flex: "1 1 280px", padding: "14px 18px", borderRadius: 999, border: "1px solid #898433",
fontSize: 17, fontFamily: "inherit", background: "#e5deca", color: "#282c15" },
  select: { padding: "14px 18px", borderRadius: 999, border: "1px solid #898433", fontSize: 17, fontFamily:
"inherit", background: "#e5deca", color: "#282c15" },
  count: { fontSize: 17, color: "#45441a", marginBottom: 24 },
  empty: { color: "#45441a", fontSize: 18, padding: 60, textAlign: "center", background: "#e5deca",
borderRadius: 16 },
  grid: {
    display: "grid",

```

```
gridTemplateColumns: "repeat(auto-fill, minmax(340px, 1fr))",
gap: 28,
},
card: {
  borderRadius: 18,
  padding: 32,
  boxShadow: "0 4px 12px rgba(40, 44, 21, 0.2)",
  border: "1px solid rgba(40, 44, 21, 0.15)",
},
cardTop: { display: "flex", justifyContent: "space-between", alignItems: "center", marginBottom: 16 },
tag: { fontSize: 13, textTransform: "uppercase", letterSpacing: 0.5, padding: "4px 10px", borderRadius: 6,
fontWeight: 700 },
outOfStock: { fontSize: 13, fontWeight: 700, color: "#e5deca", background: "#282c15", padding: "4px 10px",
borderRadius: 6 },
name: { margin: "0 0 10px", fontSize: 24, color: "#282c15" },
price: { margin: "0 0 8px", fontSize: 28, fontWeight: 700 },
qty: { margin: 0, fontSize: 16, color: "#45441a" },
};
```

Output:

Product Catalog

Search, filter, and browse the store

Search products by name...

All

Any price

Showing 16 products

ELECTRONICS

Wireless Headphones

₹2,499

12 in stock

ELECTRONICS

Bluetooth Speaker

₹1,799

Currently unavailable

ELECTRONICS

Smart Watch

₹3,999

5 in stock

CLOTHING

Cotton T-Shirt

₹599

30 in stock

CLOTHING

Denim Jacket

₹1,999

Currently unavailable

CLOTHING

Running Shoes

₹2,799

8 in stock

GROCERY

Basmati Rice 5kg

₹549

40 in stock

GROCERY

Olive Oil 1L

₹899

15 in stock

GROCERY

Almonds 500g

₹649

Currently unavailable

ELECTRONICS

Laptop Stand

₹1,299

20 in stock

ELECTRONICS

Mechanical Keyboard

₹3,499

7 in stock

CLOTHING

Wool Sweater

₹2,199

14 in stock

CLOTHING

Leather Belt

₹899

Currently unavailable

CLOTHING

Canvas Sneakers

₹1,699

22 in stock

GROCERY

Green Tea 250g

₹349

50 in stock

Product Catalog

Search, filter, and browse the store

Search products by name...

Clothing

₹1000 - ₹2500

Showing 3 products

CLOTHING

Denim Jacket

₹1,999

Currently unavailable

CLOTHING

Wool Sweater

₹2,199

14 in stock

CLOTHING

Canvas Sneakers

₹1,699

22 in stock

Product Catalog

Search, filter, and browse the store

Green Tea

All

Any price

Showing 1 product

GROCERY

Green Tea 250g

₹349

50 in stock

Conclusion:

Building the product catalog showed how managing state with `useState` lets the interface respond instantly to user input, updating results as search terms and filters change. Chaining `filter()` for the search and category conditions, `map()` for rendering, and conditional logic for stock status and product count produced a clean, fully interactive shopping experience.